

Utsav Khairnar

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Experience

AI Engineer Intern | Alma (Memfold AI), backed by Peak XV

Jun 2026 -- Present

Bangalore, On-site

- Replaced the computer-use agent's 72k-character LLM planner with a one-pass classifier over **27 tools**: task time **33.2 s** → **19.2 s**, p50 decision latency **765** → **388 ms**, accuracy **94%** → **97%**, **5k fewer tokens** per turn
- Solved fast/slow model sync: a fast model acts on screen while the main model writes its plan. Built the sync layer that skips repeated steps, drops plans made stale by wrong steps, and verifies every send or delete before it runs. Fastest end-to-end task **27 s** → **19 s**
- Built long-horizon memory for multi-step tasks in creative and CAD apps: working context in memory, task state checkpointed to disk, so tasks survive context-window limits
- Built the agent's CLI (**public launch Nov 2026**); earlier shipped **14 voice-triggered** computer-use workflows across browser, productivity, and creative apps
- Ported the multi-agent layer into a dictation app as a standalone **Rust** binary, optimized for speed and disk footprint

ML Engineer Intern | Reliance Industries Ltd.

Dec 2025 -- Mar 2026

Navi Mumbai, On-site

- Built the PDF-parsing microservice (OCR + local LLM) for the internal Letter-of-Credit scrutiny tool: **72 s** → **23 s** per document, **3,000+ documents a day** for **230 sales executives** across India
- Salt production forecasting (XGBoost) on **10 years** of data across **6 regions**: **27% more accurate** than baseline, served via FastAPI with per-region dashboards and alerts for production planning

Full-Stack Developer (Freelance) | Brydgetwork

Apr 2026 -- May 2026

New Delhi, Remote

- Co-built a donation platform sized for **4k monthly active users**: Next.js, JWT auth, payment gateway
- Owned deployment: Dockerized on a VPS behind Nginx (reverse proxy, SSL termination)

Projects

carmen: AI-written GPU kernels, proven by a judge

github.com/utsav1033/carmen

- An LLM writes Apple Metal kernels; a deterministic judge proves them against a float64 answer key on hundreds of inputs (incl. secret draws) and times them against MLX. Solo, **6 days** to a real-model result
- Fused 4-bit MLP kernels: **8 MLX kernels** → **2** per layer of Qwen 2.5 0.5B, decode **3% faster** (95% CI 2--5%, 30 paired runs) with identical output; up to **1.82×** MLX on single kernels
- Judge caught **75/75** planted bugs vs **40/75** for a KernelBench-style one-shape check; it also exposed 2 false speedups, fixed by rebuilding the benchmark

FlowGap: finds what a voice AI agent can't handle

github.com/utsav1033/flowgap

- Reads a voice agent's call transcripts, finds the caller intents its flow has no node for (calls that quietly end in a human transfer), and drafts the fix
- Offline pipeline: MiniLM embeddings, UMAP + HDBSCAN intent discovery, semantic matching to the declared flow; React Flow view of intended vs actual vs patched flow
- On **298 transcripts**, found **49%** of calls hit an unhandled intent; UMAP cut clustering noise **54.7%** → **3.7%**; gap intents recovered at **90--100% purity**

Agent-Mark: evaluation harness for black-box agents

github.com/utsav1033/agent-mark

- Traces every tool call (arguments, result, latency, tokens) and scores the run on **5 dimensions**: LLM-judged task success, programmatic tool correctness and efficiency, latency, cost
- Shipped 2 adapters, incl. Composio's open-source agent Trustclaw benchmarked on a live Gmail inbox

Skills

Languages Python, Rust, Swift, TypeScript, Java, C++, SQL

AI / Inference Computer-use agents, multi-agent systems, Metal GPU kernels, MLX, vLLM, llama.cpp, speculative decoding, prompt caching, LoRA

Backend FastAPI, Spring Boot, Next.js, Angular, microservices

ML PyTorch, TensorFlow, NumPy, scikit-learn, XGBoost, OCR

Databases SQLite, Postgres, MySQL, Redis

Infra Linux, Docker, AWS, Azure DevOps, GitHub Actions, Nginx, CI/CD

Education

B.Tech in Computer Engineering

2022 -- 2026

Pillai College of Engineering, University of Mumbai

Publications

Akshara: Contextual Transliteration of Brahmi Using Deep Learning

ieeexplore.ieee.org/document/11437445

ICNTE 2026, IEEE

- Brahmi → Devanagari → English: OCR, a deep learning transliterator trained on **22,000+** samples of **272 characters**, and a Bi-LSTM translator. Presented Jan 2026, published in IEEE Xplore